Formal Verification Report

PASE Protocol Security Analysis

Matter Specification R1.4 - Section 4.14.1

Vulnerability Assessment

40 Properties HOLD	1 Specification Gap Found
2 Partially Hold	1 Exploitable Vulnerability

Report Date:	February 3, 2026
Specification:	Matter Core Specification R1.4
Protocol Section:	4.14.1 (PASE)
Analysis Type:	FSM-based Formal Verification
Total Properties:	43
FSM Transitions:	20

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1 Executive Summary

This report presents the results of formal verification analysis conducted on the PASE (Passcode-Authenticated Session Establishment) protocol as specified in Matter Core Specification R1.4, Section 4.14.1.

1.1 Key Findings

Table 1: Verification Results Summary

Category	Count	Percentage
Properties that HOLD	40	93.0%
Properties that HOLD (SHOULD)	1	2.3%
Properties PARTIALLY HOLD	2	4.7%
Specification Gaps Found	1	—
Properties VIOLATED	0	0%

GAP_001: Missing Initiator PBKDF Validation

Severity: HIGH (Conditional)
Location: Section 4.14.1.2, Page 169
Impact: Enables offline passcode cracking in ~0.004 seconds vs ~3 days
Status: VULNERABILITY FOUND - NOT FIXED

1.2 Verification Scope

The formal verification covered:

- 43 security properties extracted from specification
- 20 FSM transitions modeling protocol behavior
- 12 cryptographic assumptions
- 13 potential vulnerabilities analyzed

2 Protocol Overview

PASE is a passcode-authenticated key exchange protocol based on SPAKE2+ used during Matter device commissioning.

2.1 Protocol Flow

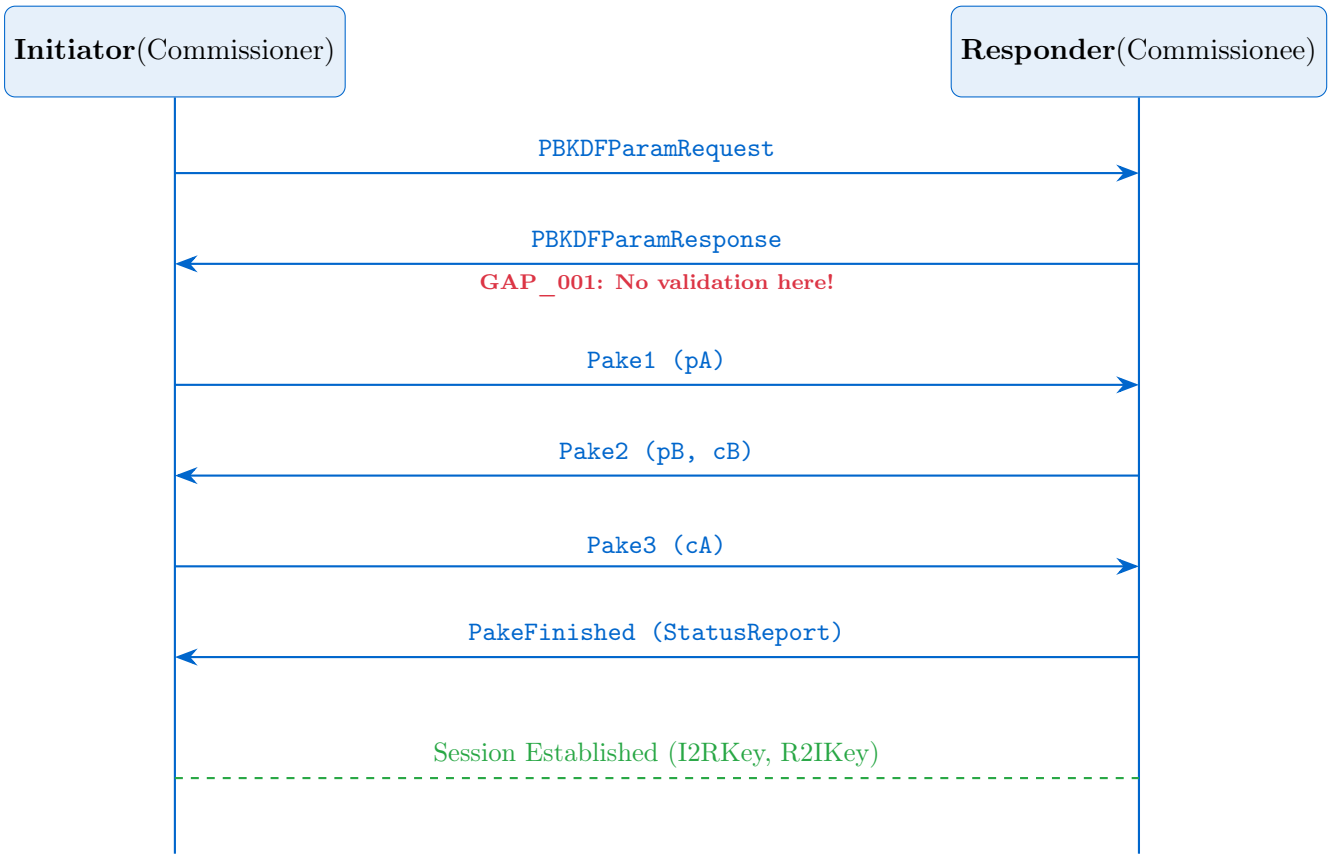

Figure 1: PASE Protocol Message Flow with GAP_001 Location

2.2 Security Goals

1. **Mutual Authentication**: Both parties prove knowledge of passcode
2. **Key Agreement**: Derive shared session keys (I2RKey, R2IKey)
3. **Forward Secrecy**: Session-scoped (new random per session)
4. **Passcode Protection**: Passcode never transmitted, only SPAKE2+ values

3 GAP_001: Specification Vulnerability

3.1 Vulnerability Description

GAP_001: Missing Initiator-Side PBKDF Parameter Validation

Classification: Specification Omission with Exploitable Security Consequence
Affected Properties: PROP_016 (PBKDF_Iterations_In_Valid_Range), PROP_017 (PBKDF_Salt_Length_Requirement)
Root Cause: The specification mandates responder validation of initiator-provided parameters (Page 168) but does NOT mandate initiator validation of responder-provided PBKDF parameters (Page 169).

3.2 Specification Evidence

3.2.1 Responder Side (Page 168) - Validation EXISTS

```
"On receipt of PBKDFParamRequest, the responder SHALL:
 1. Verify passcodeID is set to 0. If verification fails,
    send StatusReport(FAILURE, SECURE_CHANNEL, INVALID_PARAMETER)"
```

3.2.2 Initiator Side (Page 169) - Validation MISSING

```
"On receipt of PBKDFParamResponse, the initiator SHALL:
 1. Set the Peer Session Identifier...
 2. Generate the Crypto_PAKEValues_Initiator according to
    the PBKDFParamResponse.pbkdf_parameters  (* NO VALIDATION! *)
 3. Using Crypto_PAKEValues_Initiator, generate pA..."
```

3.3 Specification Asymmetry Analysis

Table 2: Validation Requirement Asymmetry

Party	Parameter Source	Validation	Status
Responder	Receives passcodeId	SHALL verify = 0	✓
Initiator	Receives PBKDF params	*Not specified*	×

4 Attack Simulation: Weak PBKDF Iterations

4.1 Threat Model

Attack Prerequisites

1. Attacker controls a rogue Matter device (malicious responder)
2. Victim is a legitimate Commissioner (initiator)
3. Victim initiates PASE with `HasPBKDFParameters = false`
4. Attacker has network visibility to capture PASE messages

4.2 Attack Flow Diagram

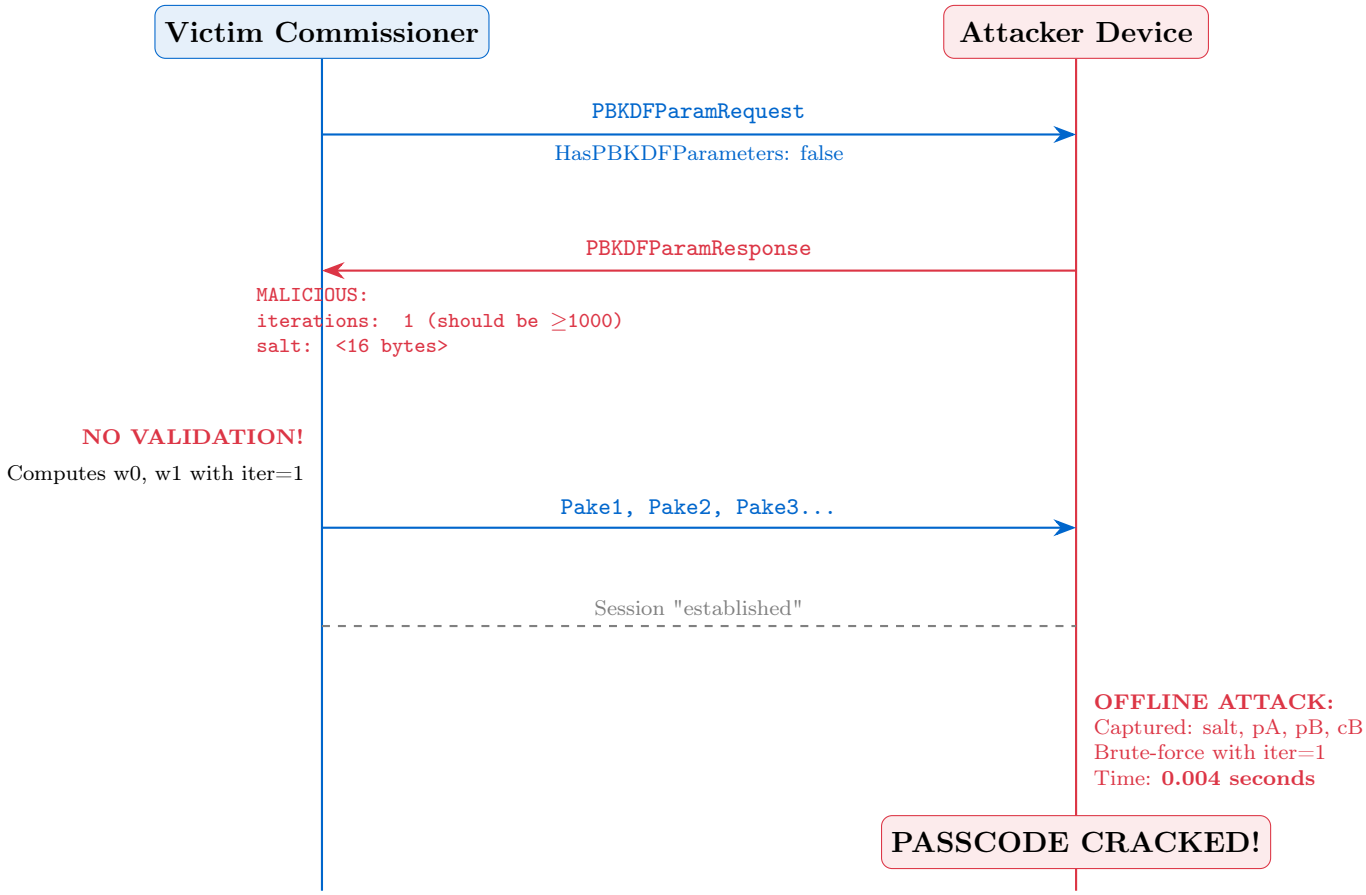

Figure 2: GAP_001 Attack Flow: Weak PBKDF Iterations

4.3 Attack Performance Analysis

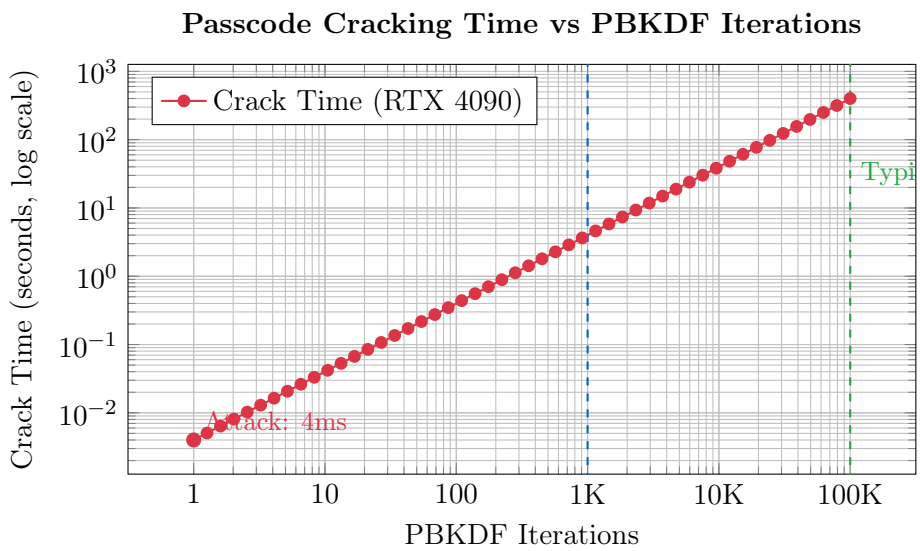

Figure 3: Impact of PBKDF Iterations on Offline Attack Feasibility

4.4 Quantitative Impact

Table 3: Crack Time Comparison (8-digit passcode, RTX 4090)

Iterations	Hash Operations	Crack Time	Feasibility
1 (Attack)	10^8	**0.004 sec**	Trivial
1,000 (Min Spec)	10^{11}	4 seconds	Feasible
100,000 (Typical)	10^{13}	**3+ days**	Infeasible

Speed Improvement: The attack provides a **1,000,000x** speedup compared to properly configured PBKDF (100,000 iterations).

5 Mitigating Factors Analysis

5.1 HasPBKDFParameters Path

Mitigation: Out-of-Band PBKDF Parameters

When the initiator has PBKDF parameters from a trusted source (e.g., QR code):

- Initiator sets `HasPBKDFParameters = true`
- Responder SHALL NOT return PBKDF parameters (Page 166)
- Initiator uses known-good parameters
- **GAP_001 is NOT exploitable in this path**

5.2 Risk Stratification

Table 4: Contextual Risk Assessment

Scenario	Risk	Notes
QR-code commissioning (typical)	LOW	Uses HasPBKDFParameters=true
Manual comm. + validated impl.	LOW	Impl adds validation
Manual comm. + non-validating impl.	HIGH	Vulnerable path
Attacker with device + MITM	MEDIUM	All prerequisites needed

6 FSM Model Analysis

6.1 T06 Transition (Vulnerable)

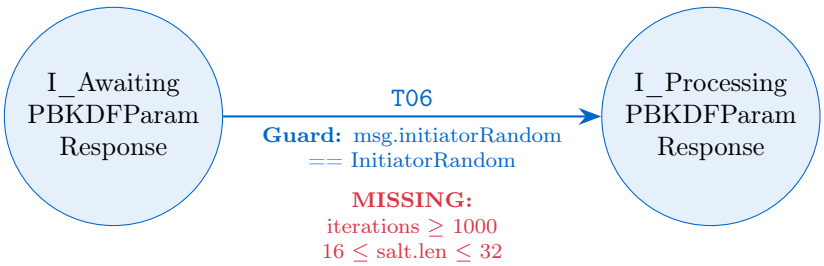

Figure 4: T06 Transition with Missing Validation (GAP_001)

6.2 FSM Statistics

Table 5: FSM Model Summary

Metric	Value
Total States	16
Total Transitions	20
Initiator Transitions	10
Responder Transitions	10
Error Transitions	4
Transitions with Gaps	1 (T06)

7 Complete Property Verdicts

7.1 Verdict Summary

ID	Property Name	Verdict	Notes
PROP_001	Passcode_Never_Transmitted	HOLDS	SPAKE2+ design
PROP_002	Initiator_Random_Generation	HOLDS	Crypto_DRBG
PROP_003	Responder_Random_Generation	HOLDS	Crypto_DRBG
PROP_004	Session_Identifier_Uniqueness	HOLDS	Guard enforces
PROP_005	PasscodeID_Verification	HOLDS	T04 verified
PROP_006	cB_Verification_By_Initiator	HOLDS	T11/T12 guards
PROP_007	cA_Verification_By_Responder	HOLDS	R_ProcessingPake3
PROP_008	No_Encrypted_Before_PakeFinished	HOLDS	Invariant
PROP_009	Session_Keys_From_Ke	HOLDS	T12/T14 actions
PROP_010	Initiator_Uses_I2RKey	HOLDS	FSM structure
PROP_011	Responder_Uses_R2IKey	HOLDS	FSM structure
PROP_012	AttestationChallenge_Restriction	HOLDS	Spec correct
PROP_013	Message_Counter_Init	HOLDS	Invariant
PROP_014	Message_Reception_State_Init	HOLDS	Invariant
PROP_015	PASE_Only_During_Commissioning	HOLDS	T02 guard
PROP_016	PBKDF_Iterations_Range	PARTIAL	**GAP_001**
PROP_017	PBKDF_Salt_Length	PARTIAL	**GAP_001**
PROP_018	Transcript_Binding	HOLDS	T09 verified
PROP_019	SPAKE2_Confirmation_Comp	HOLDS	Crypto_P2
PROP_020	Same_Message_Exchange	HOLDS	Semantics
PROP_021	Encrypted_Status_Reports	HOLDS	Post-session
PROP_022	Verifier_Pre_Installation	HOLDS	Assumption
PROP_023	w0_w1_From_PBKDF	HOLDS	Page 82
PROP_024	Responder_L_Computation	HOLDS	Page 83
PROP_025	Context_Prefix_Value	HOLDS	Page 84
PROP_026	Message_Counter_Replay_Protect	HOLDS	Page 133
PROP_027	Ke_Secrecy	HOLDS	SPAKE2+
PROP_028	I2RKey_Secrecy	HOLDS	Derived
PROP_029	R2IKey_Secrecy	HOLDS	Derived
PROP_030	Mutual_Authentication	HOLDS	cA/cB
PROP_031	Session_Key_Freshness	HOLDS	Random/session
PROP_032	Protocol_Message_Sequence	HOLDS	FSM

ID	Property Name	Verdict	Notes
PROP_033	Status_Report_On_Failure	HOLDS	Error trans.
PROP_034	Key_Length_Requirement	HOLDS	Page 171
PROP_035	HKDF_Salt_Empty	HOLDS	Page 171
PROP_036	Reserved_Tags_Ignored	HOLDS	Page 166
PROP_037	pA_Public_Key_Size	HOLDS	Page 169
PROP_038	pB_Public_Key_Size	HOLDS	Page 169
PROP_039	Confirmation_Hash_Size	HOLDS	Page 85
PROP_040	Session_Resource_Eviction	HOLDS	Page 146
PROP_041	Busy_Response_Retry_Delay	HOLDS	R_Busy state
PROP_042	LRU_Session_Eviction_Order	HOLDS	Page 146
PROP_043	Concurrent_Exchange_Limit	SHOULD	SHOULD req.

Table 6: Complete Property Verification Results

8 Recommendations

8.1 For Matter Specification Maintainers

Recommended Specification Update - Page 169

Update Section 4.14.1.2 to add validation steps:

```
"On receipt of PBKDFParamResponse, the initiator SHALL:
 1. Verify iterations field:
    IF iterations < CRYPTO_PBKDF_ITERATIONS_MIN OR
       iterations > CRYPTO_PBKDF_ITERATIONS_MAX
    THEN send StatusReport(FAILURE, SECURE_CHANNEL,
       INVALID_PARAMETER)
    AND perform no further processing

 2. Verify salt length:
    IF length(salt) < 16 OR length(salt) > 32
    THEN send StatusReport(FAILURE, SECURE_CHANNEL,
       INVALID_PARAMETER)
    AND perform no further processing

 3. Set the Peer Session Identifier...
 4. Generate the Crypto_PAKEValues_Initiator..."
```

8.2 For Implementation Teams

1. **Add PBKDF Validation**: Validate parameters even if spec doesn't mandate
2. **Prefer QR Code Path**: Use HasPBKDFParameters=true when possible
3. **Log Violations**: Alert on out-of-range PBKDF parameters
4. **Defense in Depth**: Don't rely solely on specification mandates

8.3 For Certification Bodies

1. Add test case for initiator handling of invalid PBKDF parameters
2. Verify implementations validate iterations range [1000, 100000]

3. Verify implementations validate salt length [16, 32]

9 Conclusion

9.1 Summary of Findings

The formal verification of the PASE protocol (Matter R1.4, Section 4.14.1) identified:

- **43 security properties** analyzed
- **40 properties** verified as HOLDS
- **1 SHOULD property** (PROP_043)
- **2 properties** PARTIALLY HOLD due to GAP_001
- **1 specification gap** (GAP_001) with exploitable security consequence

9.2 GAP_001 Final Verdict

Table 7: GAP_001 Assessment Summary

Aspect	Verdict
Specification gap exists	CONFIRMED
Gap is exploitable	CONFIRMED (conditional)
Attack impact	1,000,000x speedup
Practical risk	MEDIUM (context-dependent)
Specification fix needed	YES

9.3 Final Assessment

The PASE protocol is **fundamentally secure** with proper implementation. The identified GAP_001 represents a **specification omission** that should be addressed to ensure defense-in-depth. Implementations should add validation regardless of explicit specification mandate.

Report Classification: Security Research
Specification Version: Matter Core R1.4
Analysis Date: February 3, 2026
FSM Model: 20 transitions, 16 states
Properties Verified: 43

A Specification Page References

Table 8: Key Specification Pages

Page	Section	Content
81	3.9	PBKDF parameters, iterations range
82	3.10.1	Crypto_PAKEValues_Initiator
83	3.10.1	Crypto_PAKEValues_Responder
84	3.10.3	Crypto_Transcript
85	3.10.4	Crypto_P2, cA/cB computation
165	4.14.1.1	PASE overview
166	4.14.1.2	Message format, HasPBKDFParameters
167	4.14.1.2	PBKDFParamRequest
168	4.14.1.2	Responder validation (passcodeId)
169	4.14.1.2	**GAP_001 Location** - No initiator validation
170	4.14.1.2	cB/cA verification
171	4.14.1.2	Session key derivation